

Tuhin Garai

+919845779355 tuhingarai123@gmail.com Bengaluru, Karnataka, India tuhingarai.in
linkedin.com/in/tuhingarai github.com/nightcodex7 medium.com/@nightcode_x

SUMMARY

DevOps and Cloud Engineer with hands-on experience building serverless and containerized applications on AWS. Skilled in Linux administration, CI/CD automation, cloud infrastructure, and troubleshooting distributed systems, with practical exposure to Kubernetes, monitoring, and infrastructure automation.

SKILLS

DevOps & Cloud: Docker, Docker Compose, Kubernetes, Jenkins, CI/CD, AWS (EC2, VPC, S3, IAM, SNS, CloudWatch), GitHub Actions, Ansible, Terraform

Tools & Platforms: Git, GitHub, Linux

Programming/Scripting: Python, Bash, Shell Scripting

Systems & Administration: Linux Administration, System Monitoring, Troubleshooting, Log Analysis

Support & Operations: Cloud Monitoring, Incident Response, Monitoring Alerts, Root Cause Analysis, Documentation, Debugging Workflows

PROJECTS

Serverless EC2 Monitor & Control Dashboard,

Real-Time EC2 Instance Monitoring and Automation using AWS Lambda, API Gateway, DynamoDB, SES, and S3

- Built a serverless EC2 monitoring and automation system using AWS Lambda, API Gateway, DynamoDB, and S3; published findings in IJIRSET.
- Automated EC2 instance lifecycle management using usage-based triggers, reducing manual monitoring and shutdown effort by ~1 hour daily.
- Configured AWS-based monitoring, alerting, and scalable infrastructure workflows to improve observability, reliability, response time, and operational availability.
- Documented deployment architecture, workflows, and operational processes.

Zero Downtime Application Deployment using Kubernetes & AWS

- Designed a highly available cloud-native deployment architecture on Amazon EKS with multi-region failover using Route 53 health checks.
- Implemented Kubernetes rolling updates, readiness/liveness probes, and Horizontal Pod Autoscaler (HPA) for achieving zero-downtime deployments and self-healing recovery across simulated high-traffic scenarios.
- Integrated Prometheus and Grafana dashboards alongside CloudWatch for end-to-end observability of system health and scaling behavior.
- Configured warm standby disaster recovery with automated failover, reducing projected RTO by ~40% compared to cold standby baseline.

ACHIEVEMENTS

Android Open Source & Custom ROM Communities, Custom ROM Tester

06/2017 – Present

- Tested and validated custom Android ROMs (crDroid, LineageOS, AOSP, VoltageOS, and others) across 5+ devices including Nothing Phone 2, Xiaomi, and Poco series.
- Identified and reported bugs, system instability, and performance bottlenecks; captured diagnostic logs (logcat) to assist developers in debugging and optimization.
- Performed device flashing, recovery, and restoration via fastboot, TWRP, OrangeFox, and EDL mode across multiple device architectures.
- Worked with Magisk, KernelSU, and LSPosed for root configuration and system-level customization testing.

PUBLICATIONS

Published Research Paper: Build a Serverless EC2 Monitor & Control Dashboard with AWS Lambda, API Gateway, and S3,
International Journal of Innovative Research in Science, Engineering and Technology (IJIRSET)

CERTIFICATES

Ethical Hacking from National Institute of Electronics & Information Technology

National Institute of Electronics & Information Technology (NIELIT), Gorakhpur (2021)

Introduction to Industry 4.0 and Industrial Internet of Things

National Programme on Technology Enhanced Learning (NPTEL), IIT Kharagpur (2022)

EDUCATION

Masters in Computer Application (MCA), Jain University

Specialization in Storage & Cloud Technology (Major), AI & ML (Minor)

2024 – Present | Bengaluru, Karnataka

Bachelors in Computer Application (BCA), Chandigarh University

2021 – 2024 | Mohali, Punjab